SIEMENS

MAGNETOM

Troubleshooting Guide
System

Host / Imager

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR
from Impact/Expert
from Vision

Answers for life.

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1.1 General

This part of the TSG deals with problems regarding the Host and Imager. Strategy and procedures provide the necessary informations to maintain the Host and Imager.

Make sure to read the common System Strategy before you continue with Host / Imager Strategy.

1.1.1 Definition of fonts used in the documentation

command ↲	All input by the user is indicated in bold and followed by a carriage return sign.

1.2 Host strategy

Remember to return to the strategy page in use, after performing a partial repair/procedure action.				
No	**Test Step**	**Status**	**Service Action**	**Reference**
1.	System boot	Boot hangs	Check the BIOS settings	Refer to (→ BIOS / Symbios settings (NT only) / Page 10) or (→ BIOS / Symbios settings (XP only) / Page 10)
2.	Host performance	syngo MR start-up	Check performance during syngo MR start-up	Refer to (→ CPU and Memory load / Page 10)
3.	Check hard disks	Hard-disk error/s in event log	Login as administrator Fix file system & recover bad sectors	Refer to "(→ Fix filesys. & recover bad sectors (NT only) / Page 12)
4.	Check dongle id	No Licenses available	Read dongle ID, check if plugged	Refer to (→ Check Dongle ID (NT and XP) / Page 13) or Refer to (→ Dongle test (XP only) / Page 13)
5.	Monitor output	No output at LCD Monitor	Connect the LCD monitor to your service Laptop, if still defect, replace LCD monitor	Refer to (→ No output at LCD Monitor / Page 14)
		Monitor output when connected to the Laptop	Replace host (most likely the graphic PCB is defect)	Refer to (→ No output at LCD Monitor / Page 14)
		Color shading	Perform the check of color shading	Refer to (→ Color shading / Page 14)

Remember to return to the strategy page in use, after performing a partial repair/procedure action.				
No	**Test Step**	**Status**	**Service Action**	**Reference**
6.	In-Room MRC	In-room mrc suffers from any defects	In-room MRC Troubleshooting	Refer to (→ In-Room MRC / Page 14)
			Protective conductor measurement	Refer to (→ Protective conductor measurement / Page 15)
7.	MOD	No access to optional MOD drive	Check configuration of the MOD, if all correct Replace MOD	(→ Software Installation syngo MR A30 (NUMARIS/4VA30A) / MR-000.816.26), chapter "Configuration",

1.3 Imager Strategy

In most cases the first question coming up is: “Is the problem caused by hardware or software”!

Reinstalling the Imager software will answer this question quite reliable. but is relatively time consuming

The following chapter will assist you to troubleshoot problems that concerns the Imager.

Remember to return to the strategy page in use, after performing a partial repair/procedure action.				
No	**Test Step**	**Status**	**Service Action**	**Reference**
1.	Imager tests	Imager not ready	Imager tests (Service level 4 or higher, full access is needed) Shutdown the complete system, switch off the power, re-boot the system.	Refer to (→ Imager tests (NT only) / Page 21) or Refer to (→ Imager tests (XP only) / Page 27)
2.	Communication / configuration	All Imager tests failed MRC has no communication to the Imager. Error message: “The connection to a remote system broke or could not be established"	Check low-level connectivity from MRC to Imager. (Service level 7, full access is needed)	Refer to (→ Network connectivity from MRC to MPCU/Imager / MR-000.840.11)
		Ping worked to MPCU, ping failed to Imager and MPCU	Check configuration data of the Measurement, Network	Refer to (→ Physical addresses of MPCU and Imager (NT only) / MR-000.840.11)
3.	Access to NT platform of the Imager	Imager not ready	**Prerequisite:** The complete system has been shutdown, powered off, and rebooted. Use the VNC viewer to access the Imager (Only possible at customer site)	Refer to (→ Imager not ready / Page 22)
4.	Check Raw disk/ disks	Imager test 2,3, and 4 failed	**Prerequisite:** Monitor and keyboard connected to Imager or access via VNC viewer Check pixel disks (Scan for and attempt recovery of bad sectors)	Refer to (→ Check Raw disk/disks (NT only) / Page 25)
5.	Check System disk	All Imager tests failed	**Prerequisite:** Monitor and keyboard connected to Imager **or** access via VNC viewer Check system disk	Refer to (→ Check system disk / Page 24)

Remember to return to the strategy page in use, after performing a partial repair/procedure action.				
No	**Test Step**	**Status**	**Service Action**	**Reference**
6.	Format Raw disk/ disks	Raw disks are replaced	**Prerequisite:** Monitor and keyboard connected to Imager **or** access via VNC viewer	Refer to (→ Format Raw disks (NT only) / Page 26)
7.	PCI-Receiver	PCI-Receiver is replaced, Imager still fails	**Prerequisite:** Monitor and keyboard connected to Imager **or** access via VNC viewer Check access to PCI-Receiver from Imager Operating system	Refer to (→ PCI-Receiver (NT only) / Page 28)

2.1 Host Procedures

Following part provides information about procedures to be performed when replacing FRU's, boot messages and host performance.

This document covers Windows **NT** and Windows **XP** based syngo software. Select appropriate instructions depending on installed software version.

2.1.1 BIOS / Symbios settings (NT only)

Regarding BIOS / Symbios settings, refer to (→ Software Installation syngo MR A30 (NUMARIS/4VA30A) / MR-000.816.26)

In case that the host (MRC / MRSC) is not booting, the BIOS settings might have changed e.g. due to a transient in the on site power.

2.1.1.1 How to enter the BIOS settings

Output	Input
Press F2 to enter SETUP	Press **F2**

2.1.1.2 How to enter the Symbios

Output	Input
Press Ctl-C to enter Symbios	After power-on, press **Ctrl-C** simultaneously

2.1.2 BIOS / Symbios settings (XP only)

Check the BIOS settings (Install BIOS settings from BIOS CD)

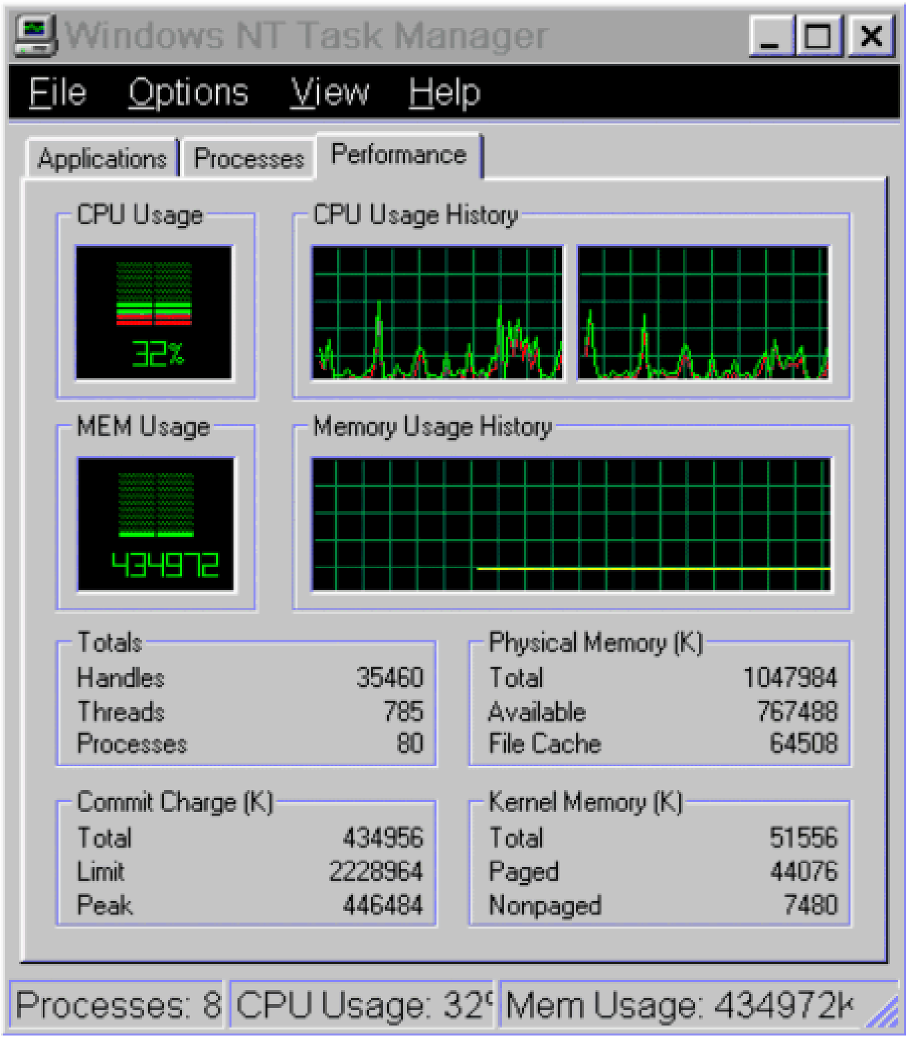
Refer to TDOC: "Install Software", valid for installed software version.

2.1.3 CPU and Memory load

To see the CPU Memory load during syngo MR start-up. Press keys **Ctrl-Shift-Esc** simultaneously click the tab **Performance**

Example, CPU and Memory load during application start-up (XP based systems has an additional tab card Network)

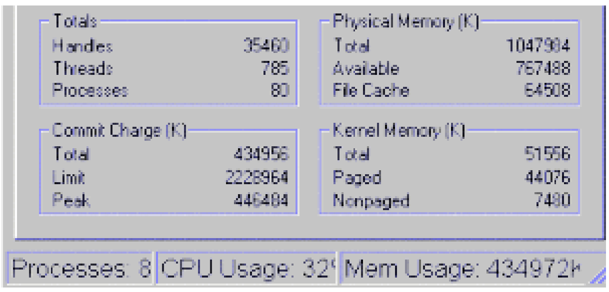
Fig. 1: NT performance monitor CPU and 512 MB RAM



2.1.3.1 Totals

Handles	The total number of handles currently opened by all processes.
Threads	A thread is the object that executes instructions. Each running process has at least one thread.
Processes	Currently number of processes.

Fig. 2: NT performance monitor



Commit Charge

A commit is the final step in the successful completion of a previously started transactions in a distributed computing system.

Physical Memory

Total usage of physical memory.

Kernel Memory

The kernel is the fundamental center of a computer operating system, it provides basic services for all other parts of the operating system.

2.1.4 Fix filesys. & recover bad sectors (NT only)

Perform Automatic fix file system errors and Scan for and attempt recovery of bad sectors.

- Reboot the system
- Login as administrator

Output	Input	Note
Windows NT	Click **Start,** select **Programs, Administrative Tools (Common)**, **Disk Administrator**	
	Click with the right mouse key at the Disk **c:** Select **Properties**	
	Click at the tab **Tools** Select **Error-checking** Click **Check Now**	The Automatic fix of file system errors can't be performed at the system disk (system partition).
	Select the check boxes: **Scan for and attempt recovery of bad sectors** Click **OK**	
Disk Check Complete		If no errors where found the message "Disk Check Complete" is displayed.
Repeat the following procedure for the Disk(s)/ Partitions **D** and **E**		
	Click with the right mouse key at the Disk / partition to be checked Select **Properties**	
	Click at the tab **Tools** Select **Error-checking** Click **Check Now**	

Output	Input	Note
	Select the check boxes: **Automatic fix filesystem errors**and**Scan for and attempt recovery of bad sectors** Click **OK**	
Disk Check Complete		If no errors where found the message "Disk Check Compleate " is displayed

- If the error-checking tool detects errors, replace the defective disk.

2.1.5 Check Dongle ID (NT and XP)

To check the functionality of the dongle, use the following procedure to read the HW dongle ID:

- Log in as user administrator, **Start**, **Run**, **cmd**
- Compare the result in the cmd tool with the dongle ID labeled at the HW dongle.**t**

Output	Input	Note
	lmutil lmhostid -flexid ø	
lmutil - Copyright (C) 1989-1998 Globetrotter Software, Inc. The FLEXlm host ID of this machine is "FLEXID=X-XXXXXXXX"		Expected result if dongle is functioning A unique HEX code will be displayed

If the manual test is successful, the dongle could still be defective and produce sporadic read errors.

2.1.6 Dongle test (XP only)

You can perform the dongle test within "Test Tools" in the Service platform. This test performs multiple reads of the dongle ID (dongle stress test).

- Start the local Service platform via Configuration, **Test Tools**, set the checkboxes "**Host**" and "**Dongle**", click "**GO**"

For detailed information regarding "Test Tools", click Help at the top right in the Service platform.

2.1.7 No output at LCD Monitor

2.1.7.1 Connect LCD monitor to the service PC

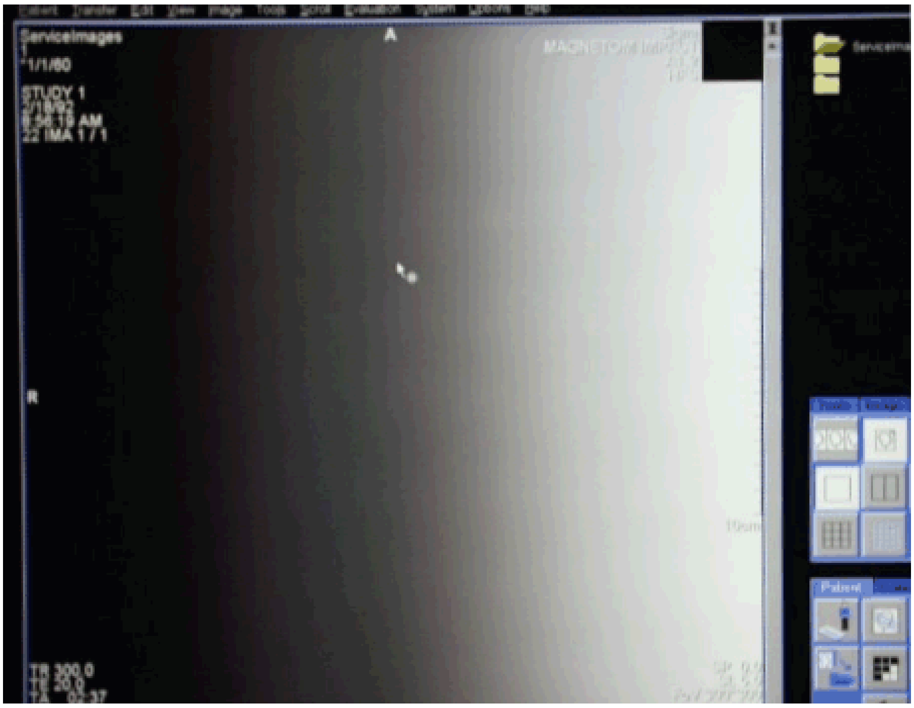
To isolate if the problem is caused by a defect monitor or missing output from the graphic PCB, connect the input cable of the LCD monitor to your service laptop.

> ℹ Remember to toggle the output of the laptop to the external Monitor.

2.1.8 Color shading

- Load the test image shown below.

Fig. 3: Test image



- Eliminate all possible sources of reflections (light off, close window...).
- The image should not show any color shading. Otherwise exchange the LCD.

2.1.9 In-Room MRC

The In-Room MRC is a option consisting of an additional LCD monitor, keyboard, and mouse located in the examination room.

2.1.9.1 Monitor

The video signal from the imager is split via the **In-Room controller in the CCA** cabinet to the LCD display and the MRC monitor. Both monitors have the same resolution and simultaneously display the same image.

2.1.9.2 Mouse

The mouse switch is used to control Numaris from the In-Room MRC. Check the function of the mouse switch at the MRC.

The LED on the mouse switch lights up to indicate that the switch is activated. If you accidentally move the mouse, the cursor of the MRC console does not respond. However, the mouse at the In-Room MRC is fully operational

2.1.9.3 Protective conductor measurement

- In case that parts has been replaced for the In-Room MRC prior switching on the system, measure the protective conductor resistance.

  Use the protective conductor meter and measure the resistance between the protective conductor connection on top of the CCA/PCA and an uncoated metal part of the In-Room MRC (e.g. screws of top cover).

WARNING

Should the resistance exceed 200mW, test the protective conductor in question for satisfactory galvanic connection.

» **In the interest of the safety of our personnel and others, the protective ground wires must be installed prior to first switching on the product/system, as well as after completing all work, before turnover to the customer, in accordance with the product documentation.**

2.1.9.4 Assignment of LED's

The LED's are located at the back side of the In-Room MRC.

Tab. 1 Overview of LED’s

LED	Function	Assignment
LED 1	Voltage regulator	5 V
LED 2	External power supply	12 V
LED 3	In-Room mouse active	PS/2
LED 4	Optional/no function, intended for space mouse active	RS232
LED 5	Video signal	RGB

2.1.9.5 Status of LED's

LED's 1, 2, and 5 must light up continuously during operation.

LED 3 only lights up when you move the mouse.

LED 5 goes off when you disconnect the video cable from the video distributor.

2.1.9.6 Error messages

Tab. 2 Overview of error messages

Error	Error message/ additional indications	Cause	Other potential causes
In-Room MRC monitor is blank, not even the test image (Siemens Logo) displays after turning on the MRC	12V status LED on the back side lights up	Monitor defective	
	12V status LED on the back side is off and power LED on the power supply is off	Power supply defective	
	12V status LED on the back side is off and power LED on the power supply lights up	Cable harness (FOC RGB) is defective or monitor (FOC receiver) is defective	
In-Room MRC monitor is blank, but test image (Siemens Logo) displays after turning on the MRC	Local monitor is blank as well	PC has power management enabled	
		PC is turned off or defective	
		Video multiplexer is turned off or defective	
		RGB connection cable PC to video multiplexer is loose or defective	
	Image present on local monitor and RGB status LED on the back side lights up	Monitor defective	
	Image present on local monitor and RGB status LED on the back side is off	Monitor in standby mode (move the In-Room Mouse to exit standby mode) or FOC cable (9 pin SUBD) at video multiplexer is loose or cable harness (FOC RGB) is defective or video multiplexer is defective	
No cursor movement on monitor after moving the local PS/2 mouse		Mouse multiplexer is off or PS/2 cable at mouse multiplexer is loose or defective	

Error	Error message/ additional indications	Cause	Other potential causes
No cursor movement on In-Room monitor after moving the In-Room PS/2 mouse	5V status LED on the back side lights up	Mouse multiplexer takes approx. 1 minute to activate the In-Room mouse after power-on (please wait)	
		PS/2 status LED on the back side lights up when moving the In-Room mouse	PS/2 cable PC to mouse multiplexer not connected or defective
		PS/2 status LED on the back side is off when moving the In-Room mouse	FOC cable at mouse multiplexer not/incorrectly connected or cable harness (FOC duplex) is defective or PS/2 mouse not initialized (disconnect and reconnect the mouse connector) or PS/2 mouse defective
	5V status LED on the back side is off	Monitor defective	
No cursor movement on the monitor after moving the local RS232 mouse		The local RS232 mouse is usually directly connected to the PC's COM port. Operating the mouse through the mouse multiplexer requires sending a special switching code to the multiplexer	
No cursor movement on In-Room monitor after moving the In-Room RS232 mouse	5V status LED on the back side is off	Monitor defective	
	5V status LED on the back side lights up	RS232 status LED on the back side lights up when moving (clicking) the In-Room mouse	RS232 cable PC to mouse multiplexer not connected or defective

Error	Error message/ additional indications	Cause	Other potential causes
		PS/2 status LED on the back side is off when moving the In-Room mouse	FOC cable at mouse multiplexer not/incorrectly connected or cable harness (FOC duplex) is defective or RS232 mouse not initialized (disconnect and reconnect the mouse connector) or RS232 mouse defective
Power LED on the power supply is off	Mains fuses on the power supply defective?		
	Mains supply lines not connected/defective?		
	Main fuse for In-Room MRC OK?		
	Secondary fuse F101 on regulator board defective? (only accessible by opening the housing)		

2.1.9.7 Protective conductor measurement

- In case that parts has been replaced for the In-Room MRC prior switching on the system, measure the protective conductor resistance.

  Use the protective conductor meter and measure the resistance between the protective conductor connection on top of the CCA/PCA and an uncoated metal part of the In-Room MRC (e.g. screws of top cover).

WARNING

Should the resistance exceed 200mW , test the protective conductor in question for satisfactory galvanic connection.

» **In the interest of the safety of our personnel and others, the protective ground wires must be installed prior to first switching on the product/system, as well as after completing all work, before turnover to the customer, in accordance with the product documentation.**

2.2 Imager Procedures

The following procedures are provided for search of Imager related errors.

2.2.1 Imager tests (NT only)

The Imager tests include communication, raw data handling, image calculation, and feedback loop Imager MMC (timing critical).

All Imager tests are Functional Tests

2.2.1.1 Prerequisites

- MMC tests are performed, completed successfully (Testtype Functional & Comm/ Status)
- Network connection from Host to Imager is available

Service level 4 or higher, (full access) is needed for following tests:

2.2.1.2 Start Imager tests

At the MRC: Select **Test Tools**, Select **Imager,** click **Go** in the service platform

Initialization

This step initializes MMC and Imager for Imager tests. If this test fails, check the network between MRC and Imager

Test 1: Data Transfer

This step checks if the fiber optics link between RX4 in the MMC to the PCI-Receiver in the Imager is OK.

In case of an error, one or more of the following units might be defective:

- RX4
- Fiber optics link between MPCU and Imager
- PCI-Receiver
- Memory in Imager

Test 2: Data acquisition

PCI-Receiver and Memory in Imager are tested. In case of an error, one or more of the following units might be defective:

- PCI-Receiver
- Memory in Imager

Test 3: Image Calculation

(Raw data source: File.)

In case of an error, one or more of the following units might be defective:

- Memory in Imager
- Hard disk for preprocessed measurement data

Test 4: Image Calculation

(Raw data source: Receiver.)

In case of an error, one or more of the following units might be defective:

- PCI-Receiver
- Memory in Imager
- Hard disk for preprocessed measurement data (Pixel disk of Imager)

Test 5: Image Calculation

(Source: Receiver.)

In case of an error, one or more of the following units might be defective:

- PCI-Receiver
- Memory in Imager
- Hard disk for preprocessed measurement data
- Network performance problem between Imager and Host

Feedback Loop Imager-MMC

Feedback Loop Imager-MMC is a performance test, a normal feedback time result is 2 to 10 ms.

Prerequisite: Imager Tests 1-5 have been completed successfully.

In case of an error, one of following connections might be defective:

- Ethernet between Imager and MMC (performance problem)
- Fiber optics link between MPCU and PCI-Receiver (performance problem)

2.2.2 Before replacing hardware

Re-boot Imager, repeat the Imager tests

> The test ‘Feedback loop Imager - MMC’ has a critically set time limit, this test might fail in 1 of 20 performed tests at Imagers without any defects!

2.2.3 Imager not ready

With the following procedure we are trying to isolate the reason for the "un-ready" state of the Imager.

Fig. 4: Imager not ready

2.2.3.1 Prerequisites

- Restart Applications has been performed in System, Control, "Image Reconstruction System" or the complete system has been shutdown and powered off, switched on and re-booted.
- Network connectivity tests has been performed to the Imager with success. Refer to (→ Network connectivity from MRC to MPCU/Imager / MR-000.840.11)

2.2.3.2 Access Windows NT of the Imager (NT only)

Following procedure is only possible to perform at the customer site: You can access the Windows NT of the Imager via the MRC **or** the Service PC with a network interface.

Start VNC viewer at Service PC

Connect your Service PC to the local Measurement system network switch, refer to (→ Network settings for service Laptop / MR-000.840.10)

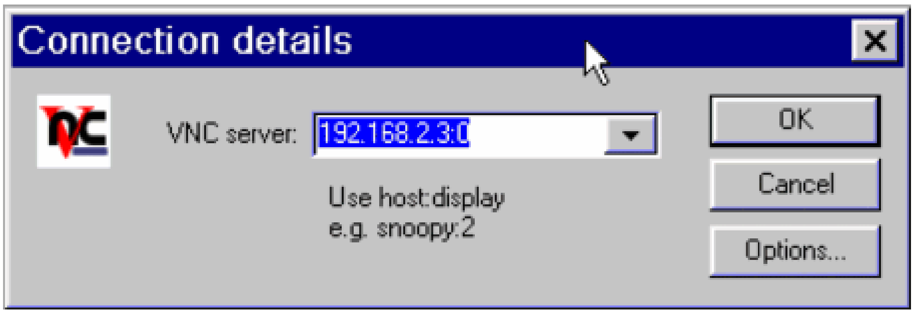
Start VNC viewer at MRC

- Start the Local service, Select **Utilities**, **Escape to OS,**

Output	Input
	c:\medcom\utils\vncviewer.exe ↲

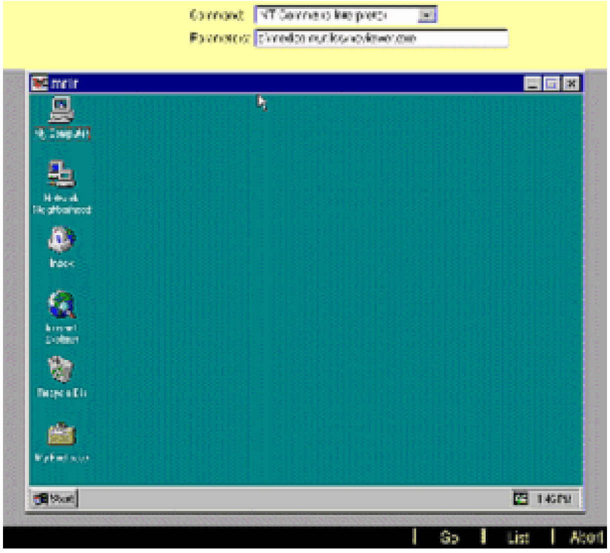
Establishing connection to the imager

Fig. 5: VNC start connection to Imager



Output	Input	Note
VNC server	**192.168.2.3:0** ↲	
Session password	**mrir** ↲	

Fig. 6: Windows NT of the Imager via VNC



2.2.4 Imager harddisks

2.2.4.1 Check system disk

- Access the Windows NT platform of the Imager, refer to (→ Access Windows NT of the Imager (NT only) / Page 23) **or** connect the Monitor, mouse, and keyboard from the Host to the Imager

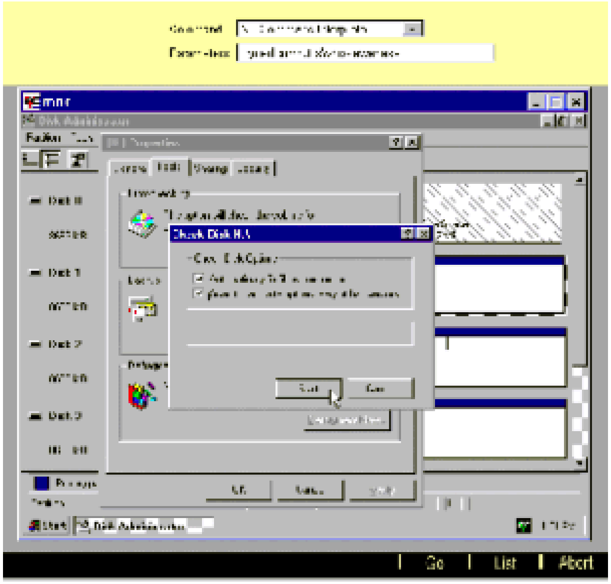
Check the system disk in Windows NT of the Imager (via VNC)

Output	Input	Note
Windows NT	Click **Start,** select **Programs, Administrative Tools (Common)**, **Disk Administrator**	
	Click with the right mouse key at the Disk **c:** Select **Properties**	
	Click at the tab **Tools** Select **Error-checking** Click **Check Now**	The Automatic fix of file system errors can't be performed at the system disk.
	Select the check boxes: **Scan for and attempt recovery of bad sectors** Click **OK**	
Disk Check Complete		If no errors where found the message "Disk Check Complete" is displayed
Repeat the following procedure for all Disk(s)/ Partitions		

2.2.4.2 Check Raw disk/disks (NT only)

Check the raw disks in Windows NT of the Imager (via VNC)

Fig. 7: Checking disk raw disk F ....



- If the error-checking tool detects errors, replace the defective disk.

Output	Input	Note
Windows NT of the Imager via VNC	Click **Start,** select **Programs, Administrative Tools (Common), Disk Administrator**	
	Click with the right mouse key at the Disk **F:** Select **Properties**	For high performance systems test the disks (**G**,**H**, and **I**) as well.
	Click at the tab **Tools** Select **Error-checking** Click **Check Now**	
	Select the check boxes: **Automatic fix filesystem errors**and**Scan for and attempt recovery of bad sectors** Click **OK**	
Disk Check Complete		If no errors where found the message "Disk Check Complete" is displayed

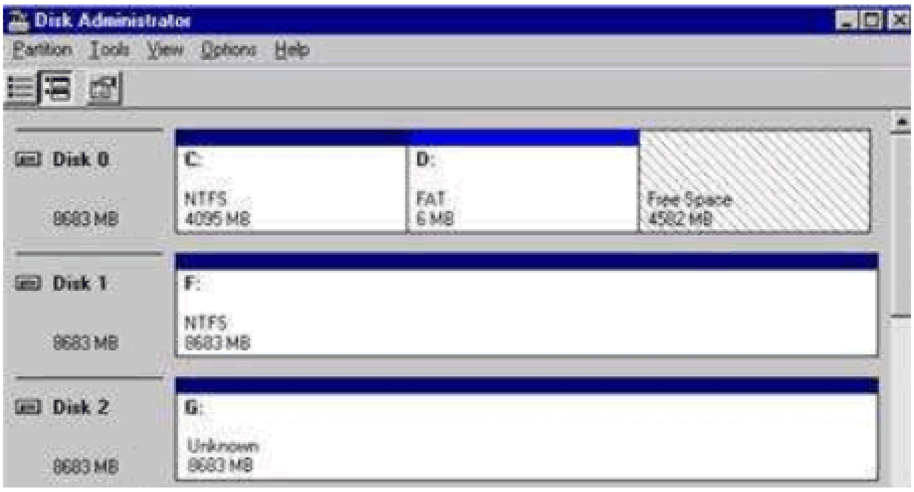
- Repeat the above "Error-checking" procedure for all raw disks at high end systems (4 x Raw-disks).

2.2.4.3 Format Raw disks (NT only)

- Access the Windows NT platform of the Imager, refer to (→ Access Windows NT of the Imager (NT only) / Page 23) **or** connect the Monitor, mouse, and keyboard from the Host to the Imager
- Click START ,Programs, Administrative Tools (Common), Disk Administrator

Output	Input	Note
	Confirm the dialogue box First Disk configuration with **OK** "Do you want to write a signature on disk ... so that Administrator can access the drive, answer **Yes** for each Raw disk	
	Do not perform any changes for **Disk 0**	(The system disk is partitioned in two parts C: and D:)

Fig. 8: Formatting, partitioning, and creating NTFS file system



- Check/Set maximum partition size for Raw disk (Raw disk 1-4 high end systems)

Output	Input	Note
	Click with the right mouse key in the gray lined field, Select **Create** Click **OK** , Click with the right mouse key in the white field, select **Commit changes now** Do you want to save changes, click **Yes**	Make sure that maximum size is set

Output	Input	Note
Format Raw Disk(s). 1 disk low end (1-4 disks high end systems)		
	Click with the right mouse key in the white field, select **Format** Select **NTFS** Select **Quick Forma**t by setting the check-box Click **Start** Click **OK** Click **OK**	Use the file system type **NTFS** for Disk1 (Disk 1-4 high end systems)

- Re-boot the system

The Imager will reboot twice following the installation.

The initialization of each pixel disk takes approximately ~5 minutes

When the initialization is finished the raw disks will contain fillowing files: (You can use the VNC to access the Imager)

F:

MattauchBuffer1.dat

rawexchange.dat

G:

MattauchBuffer2.dat

rawexchange.dat

H:

rawexchange.dat

J:

rawexchange.dat

In case you can’t repair disk 2-4 with format on a high end system you can re-configure at the MRC in Configuration, Measurement, MRIR “NUMARIS/4 MRIR settings”

# of Harddisks to 1 as temporary solution until you get spare parts.

2.2.5 Imager tests (XP only)

Perform the "Imager" tests.

- Start the local Service platform Test Tools; select the "**Imager**" checkbox at the far right; click "**GO**"

For detailed information regarding "Test Tools" and "Imager Tests", click Help at the top right in the Service platform.

After performing "Imager Tests", reboot the system

2.2.5.1 Accessing Windows XP at the Imager

To access "Windows XP" at the Imager, use "Remote Desktop Connection"

At the "MRC", simultaneously press **Ctrl - Esc**. Select **Start**, **Advanced User**; enter the advanced user password; click **OK**

Simultaneously press **Ctrl - Esc**; select **Start**, **Programs**, **Accessories**, **Communications**, **Remote Desktop Connection**; enter the computer name of the Imager: mrir, log in as user: **MRSystem**, enter the password for account "MRSystem" and click **GO**

Now you will see the "Imager" desktop on the MRC.

E.g., you can check if the image "rawexchange" drives F:\ G:\, H:\, I:\) are available and if the "rawexchange.dat" files are available on each "rawexchange" drive.

2.2.6 PCI-Receiver (NT only)

With the following procedure you can verify that the PCI-receiver board is correctly plugged in to the PCB slot.

- Access Windows NT of the Imager via the VNCVIEWER refer to (→ Access Windows NT of the Imager (NT only) / Page 23)

Output	Input	Note
Windows NT of the Imager via VNC	Click **Start,** select **Run,** enter the command **cmd**	
	mafu.cmd	
If the access is **successful** to the PCI-receiver following message is displayed: **"Putting data_top_00 in registry" or "Putting data_top_01 in registry"**		
If the access is **fails** to the PCI-receiver following message is displayed: **"An error occurred while retrieving the Ausfuehrungsstand"**		

» The common print number (MR-000.840.01) from the Troubleshooting Guide is divided into separate Component print numbers.

4 List of Hazard IDs

There are no Hazard IDs in this document.

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Siemens Healthcare Headquarters
Siemens Healthcare GmbH
Henkestrasse 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthcare

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